

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination –2025

Course: F.Y.B. Tech Engg.

Branch: Common to All branches

Semester : I

Subject Name & Subject Code: Engineering Chemistry (23UD1CHEBS102)

Max Marks: 60 M

Date: 17/01/2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Residual hardness of Hot lime soda process is _____.	1	1
	a. 2-4 ppm ☒ 15-30 ppm c. 5-7 ppm d. 10-12 ppm		
2	Cation exchange resins can be represented as _____.	1	1
	a. OH b. KH ☒ R-H ⁺ d. ROH ⁻		
3	Hardness of water is expressed in terms of _____.	1	1
	a. Kg b. ml c. cm ☒ ppm		
4	_____ corrosion occurs by the formation of pinholes in the metal.	2	1
	a. Microbial b. Galvanic ☒ Pitting d None of these		
5	As _____ increases the rate of corrosion also increase.	2	1
	☒ temperature b. purity c. base d None of these		
6	The rate of corrosion is more _____ medium.	2	1
	a. neutral b. basic c. amphoteric ☒ acidic		
7	Highest percentage of carbon present in _____ coal.	3	1
	a. Peat b. Lignite ☒ Anthracite d. Bituminous		
8	Graphite when dispersed in water is known as _____.	3	1
	☒ Aqua dag b. Oil dag c. Mineral dag ☒ Graphite		
9	Glass electrode is used to determine _____ of solution.	4	1
	☒ Conductance b. pH c. K ⁺ d. None of these.		
10	In conductometric titration, Strong Acid-Strong base titration gives _____ graph.	4	1
	a. S shape ☒ V Shape c. W shape d. None of these.		
11	In Thin Layer Chromatography, _____ is used as stationary phase.	5	1
	☒ Silica b. sodium c. Liquid d. Chloroform		
	Alumina gel		

12	IR Spectroscopy depends upon _____ principle.	5	1
	☒ a. Adsorption ☐ b. Vibrational-Rotational ☐ c. Electronic Excitation ☐ d. chemical shift		
Q. 2	Solve the following.		12
☒ A)	Discuss the disadvantages of hard water in domestic and industrial use.	1	6
B)	Define the term dissolved oxygen (DO). Explain determination of dissolved oxygen by Winkler's method.	1	6
Q.3	Solve the following.		12
☒ A)	Write a note on dry (direct chemical) corrosion.	2	6
☒ B)	Discuss cathodic protection method to minimize the rate of corrosion.	2	6
Q. 4	Solve Any Two of the following.		12
☒ A)	What is Fuel? Give the classification of Fuels.	3	6
B)	Write a note on Refining of Petroleum.	3	6
C)	What is Lubricant? Explain the type of Lubricants.	3	6
Q.5	Solve Any Two of the following.		12
A)	Define Fuel cell. Explain the working of fuel cell.	4	6
☒ B)	Explain conductometric titration with any two suitable examples.	4	6
☒ C)	Write a note on Glass Electrode.	4	6
Q. 6	Solve Any Two of the following.		12
A)	Explain instrumentation and working of Double beam UV-Visible spectrophotometer.	5	6
B)	Explain the classification of Chromatographic Technique.	5	6
C)	Describe principle and working of flame photometer.	5	6
*** End ***			